

IEEE x Knight Hacks Embedded Workshop!

Arduino IDE and Driver Installation Guide

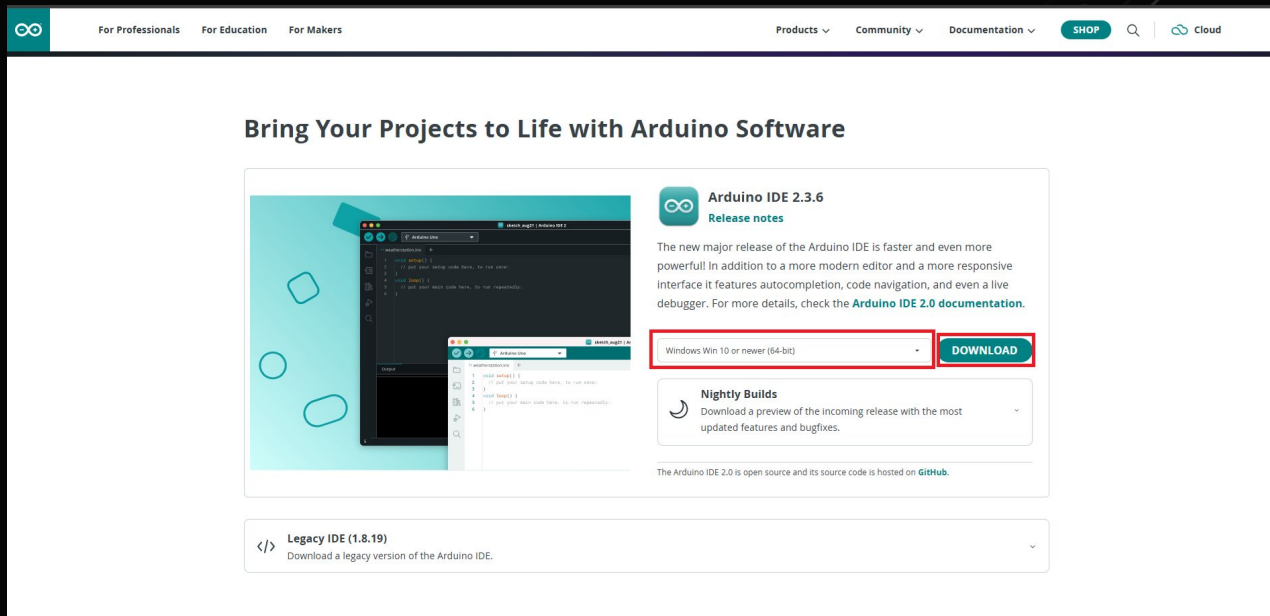


[IEEE Discord Link](https://discord.gg/hAGRkgEBZQ)



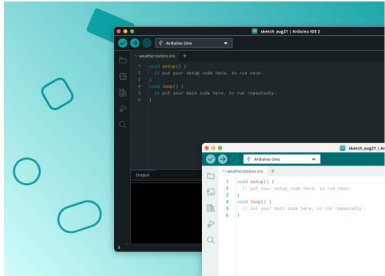
[Github Link](#)

How to Install and Configure the Arduino IDE



For Professionals For Education For Makers Products Community Documentation SHOP Cloud

Bring Your Projects to Life with Arduino Software



Arduino IDE 2.3.6 Release notes

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

Windows Win 10 or newer (64-bit) **DOWNLOAD**

Nightly Builds
Download a preview of the incoming release with the most updated features and bugfixes.

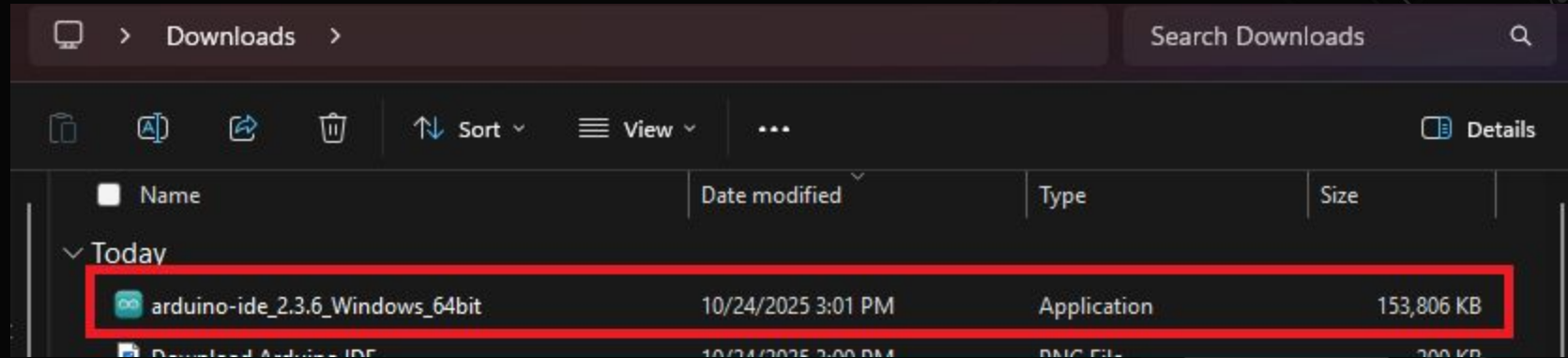
The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

Legacy IDE (1.8.19)
Download a legacy version of the Arduino IDE.

Select your OS and click “DOWNLOAD”

Link: <https://www.arduino.cc/en/software/>

How to Install and Configure the Arduino IDE continued...



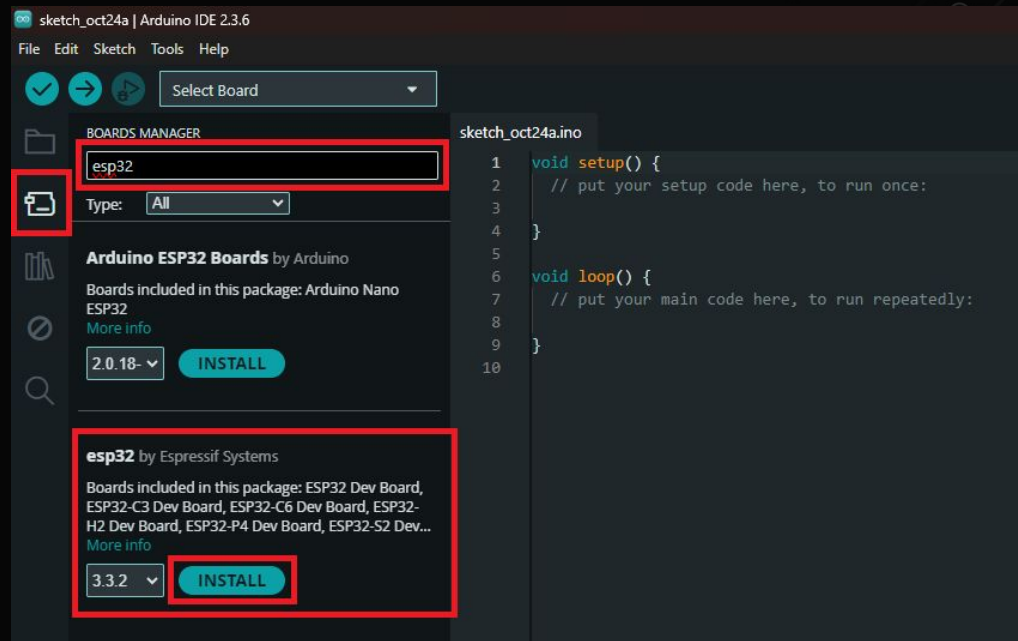
Next, run the installer!

Additional slide:

Go to File, Then Preferences, then in Additional Board Manager URL's, Paste this:

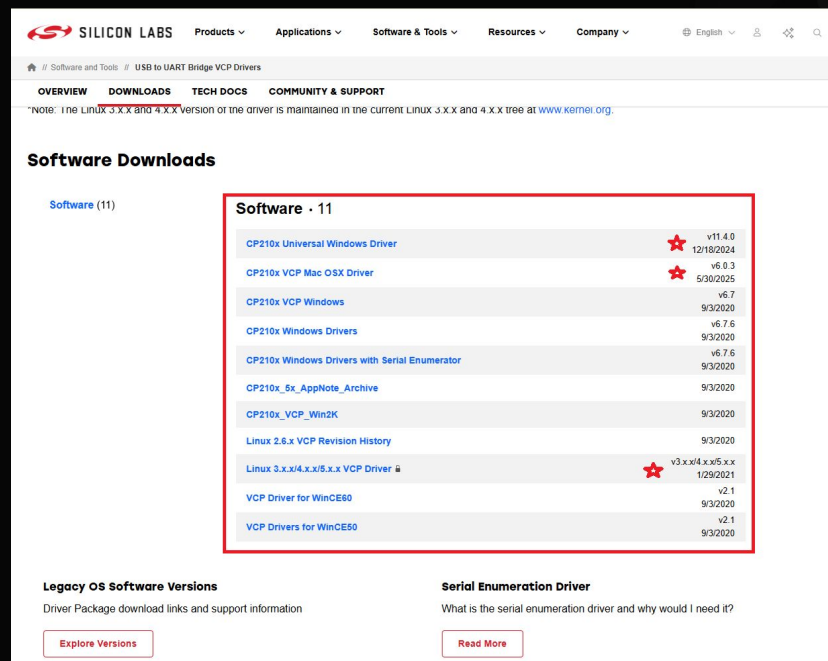
https://espressif.github.io/arduino-esp32/package_esp32_index.json

How to Install and Configure the Arduino IDE continued...



Open the IDE > Select the “Boards” tab > search “ESP32” > Install the “esp32” pack by Espressif Systems

How to Install and Configure the Drivers...



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Software and Tools # USB to UART Bridge VCP Drivers

OVERVIEW **DOWNLOADS** TECH DOCS COMMUNITY & SUPPORT

*Note: The Linux 3.x.x and 4.x.x version of the driver is maintained in the current Linux 3.x.x and 4.x.x tree at www.kernel.org.

Software Downloads

Software (11)

Software · 11

CP210x Universal Windows Driver	★	v11.4.0 12/18/2024
CP210x VCP Mac OSX Driver	★	v6.0.3 5/30/2025
CP210x VCP Windows		v6.7 9/3/2020
CP210x Windows Drivers		v6.7.6 9/3/2020
CP210x Windows Drivers with Serial Enumerator		v6.7.6 9/3/2020
CP210x_5x_AppNote_Archive		9/3/2020
CP210x_VCP_Win2K		9/3/2020
Linux 2.6.x VCP Revision History		9/3/2020
Linux 3.x.x/4.x.x/5.x.x VCP Driver	★	v3.x.x/4.x.x/5.x.x 1/29/2021
VCP Driver for WinCE60		v2.1 9/3/2020
VCP Drivers for WinCE50		v2.1 9/3/2020

Legacy OS Software Versions

Driver Package download links and support information

[Explore Versions](#)

Serial Enumeration Driver

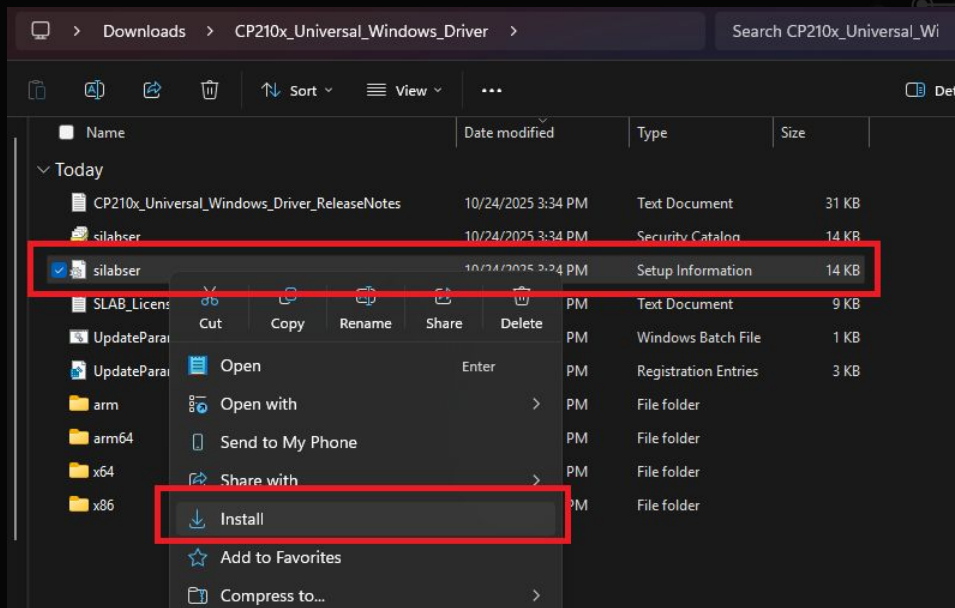
What is the serial enumeration driver and why would I need it?

[Read More](#)

Download the appropriate driver based on your OS

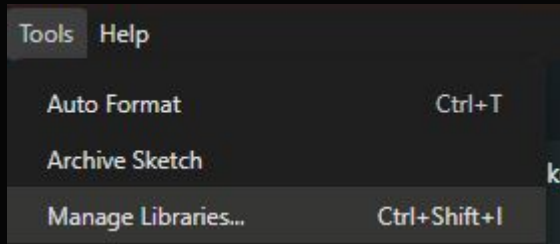
Link: <https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads>

How to Install and Configure the Drivers continued...

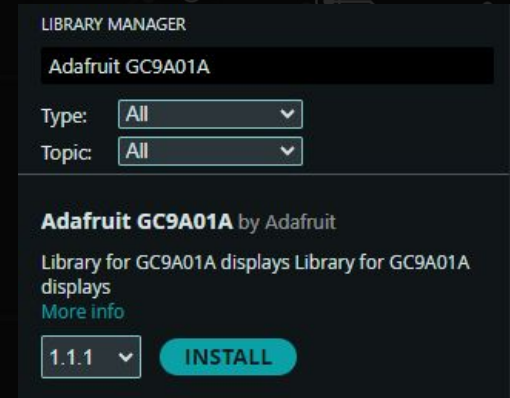


Extract the folder > Run the “silabser” .inf/setup information file installer (right-click and select “install”)

How to Install and Configure the IDE with Libraries...

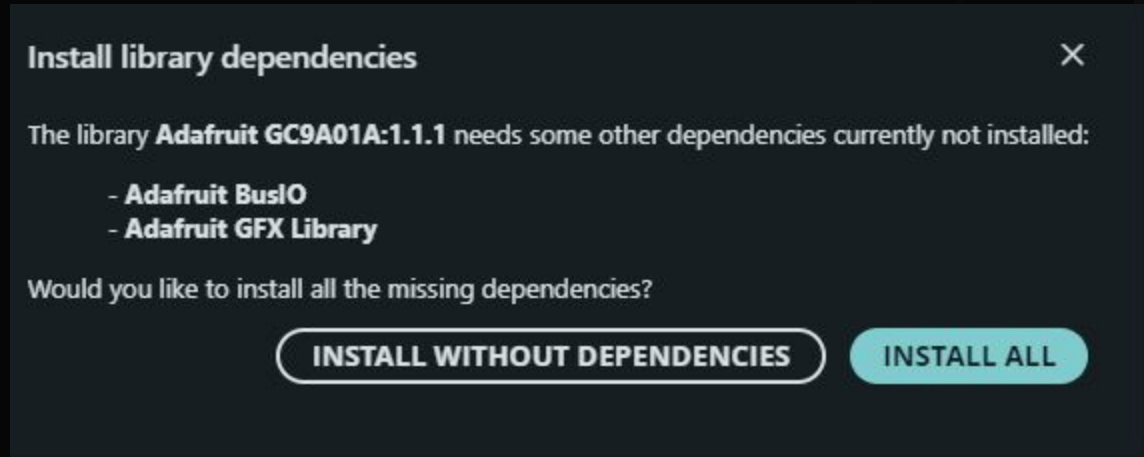


Once you load the IDE,
go to tools and go to
manage libraries.



You will see a library
manager pop up, in
the search bar, type in
"Adafruit GC9A01A"

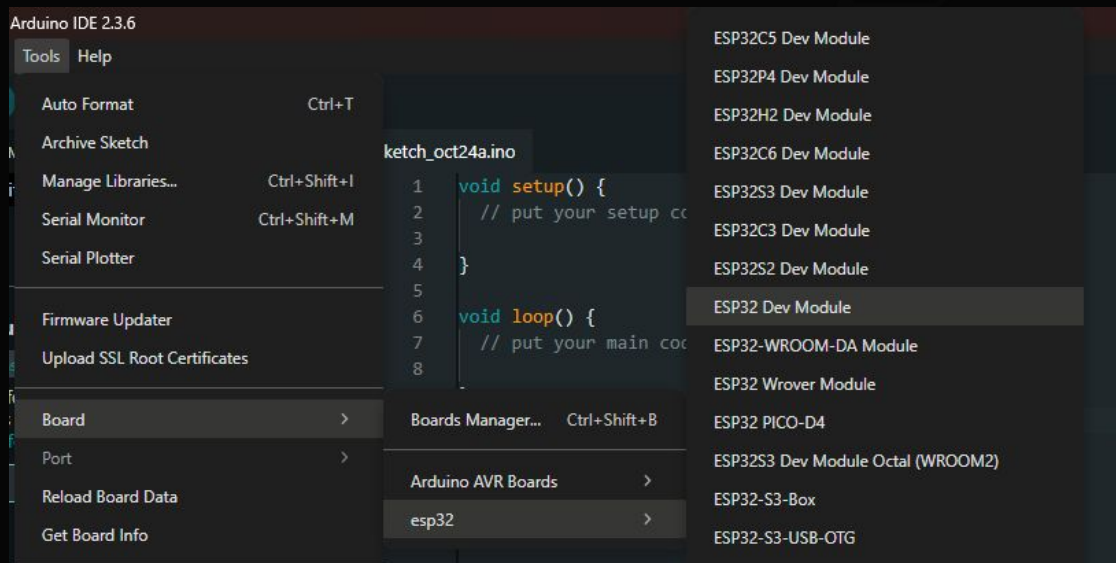
How to Install and Configure the Arduino IDE continued...



It will ask you to install 2 additional libraries, press **INSTALL ALL**.

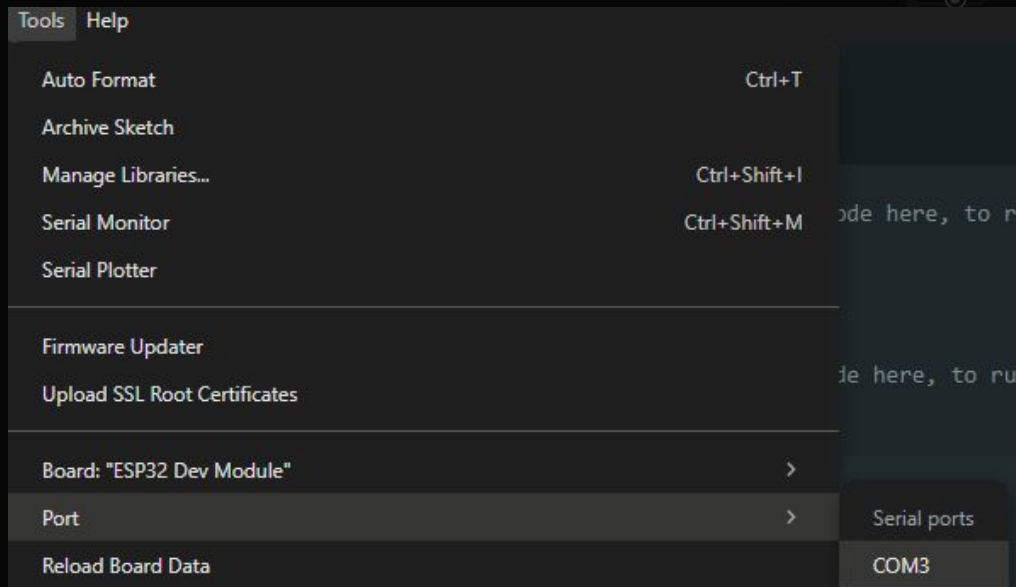
The Display in Session 2 will not work without it. :(

How to Install and Configure the Arduino IDE continued...



Go back to tools, then go the the board section, then hover over ESP32, then select “ESP32 Dev Module”

How to Install and Configure the Arduino IDE continued...



Then, once you have selected the correct board, you will see you finally have the option to select a Port. Select the COM3 Port.

And then you're done! :)

QR Codes!!!!



[Feedback Link](#)



[IEEE Discord Link](#)